BDD(Behavioral Driven Development) Study Note

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1. Definition & Core Philosophy
- **Definition**: BDD stands for **Behavior Driven Development**1.
- **Core Purpose**: To bridge the communication gap and understanding mismatch between business teams (Product Owners, Business Analysts) and technical teams (Developers, Testers).
- **Core Mechanism**: Fostering active collaboration across disciplines to build a common, shared understanding of application behavior.
- **Output**: A **Single Source of Truth**, also referred to as "**Living Documentation**", accessible and understandable to all team members and stakeholders.

2. Evolution of BDD (TLD → TDD → BDD)
- **TLD (Test-Last Development)**: Traditional approach where application code is written first, followed by test case creation and execution.
- **TDD (Test-Driven Development)**: A test-first approach where test cases are created first (which initially fail), and code is written to make those tests pass.
- **BDD (Behavior-Driven Development)**: An extension of TDD89. It retains the test-first concept but shifts the focus to describing **business behavior from the end-user's perspective** using human-readable language (such as Gherkin) to enhance collaboration across technical and non-technical roles.

3. Three Core Practices / Phases of BDD
1. **Discovery Phase**:
   - Begins with a User Story written from the end-user's perspective1012.
   - The team holds a "**Three Amigos**" meeting (Business, Developer, Tester) to discuss, explore, and propose concrete examples10more_horiz.
   - Outcome: **Agreed System Behavior**.
2. **Formulation Phase**:
   - Translates the agreed examples into business-readable and executable scenarios using specification languages like Gherkin.
   - Outcome: **Documented Business-Readable and Executable Scenarios**.
3. **Automation Phase**:
   - Converts scenarios into automated tests (Step Definitions / Glue Code) using BDD tools like Cucumber16more_horiz.
   - Automated tests guide actual code development.
   - Outcome: **Implemented Application Code and Executable Test Suites**.

4. Roles & Responsibilities
- **The "Three Amigos"**:
  - **Business Representative (BA / Product Owner)**: Defines business problems and requirements, providing user stories and business context.
  - **Developer**: Formulates technical solutions, designs software architecture, and writes implementation code1325.
  - **Tester / QA**: Defines test strategies, identifying edge cases, failure scenarios, and impacted modules13more_horiz.
- **Other Essential Roles** (BDD emphasizes whole-team collaboration):
  - **Solution Architect**: Analyzes technical requirements and decides tech stacks, environments, and architecture25.
  - **Operations / Network Teams (NetOps)**: Manages infrastructure, server deployment, and communication networks.

5. BDD vs. Traditional Development Process
- **Traditional Process**: Dev and QA are not consulted during requirements gathering; teams work in silos; testing occurs at the end of development.
- **BDD Process**: Encourages early, cross-functional collaboration via Three Amigos meetings to establish a shared single source of truth before coding begins.

6. BDD Workflow in Sprints

- **Pre-Sprint**: Requirements gathering, user story creation, Three Amigos discovery meeting, and formulating scenarios in Gherkin syntax.
- **During Sprint**: Concurrent automation test creation and feature development, followed by test execution, bug fixes, regression testing, and release.

7. Specification Syntax & Tools

- **Gherkin Syntax**: A domain-specific language readable by all team members2033. Key words include:
  - **Feature**: Name of the feature under test.
  - **Scenario**: Specific business use-case.
  - **Given**: Prerequisite state.
  - **When**: Action or event.
  - **And**: Additional conditions or actions.
  - **Then**: Expected outcome or result.
- **BDD Tools**: Tools like **Cucumber**, JBehave, and Behat map Gherkin scenarios to backend step definition code to run automated tests.